Locked or Leaked? Speaker Script

MCCJ 525 | 10-minute group presentation | 5 members × 2 slides × ~1 minute each

Tips: *Text in [square brackets] is a stage cue, not something to say aloud. Each section is about 150 words, roughly one minute at a calm pace. Practise once with a timer.*

MEMBER 1 – [Your Name]

Slide 1: Cold open: One message. One night. Five doors. (about 1 minute)

[Speak slowly. Let the phone on the slide do the work.]

It is 11:47 on a normal weeknight in Dhaka. Nusrat, a 21-year-old university student, is scrolling in bed when a message arrives from her friend Rafi: "I sent a six-digit code to your number by mistake. Can you send it to me? It's urgent."

It looks harmless. It is her friend. She types the code and hits send.

Two minutes later, her WhatsApp is no longer hers.

[Pause.]

Good morning. We are Group [__], and this is "Locked or Leaked?" Nusrat is not a real person, but everything that happens to her tonight happens to real students in Bangladesh every week.

Over the next ten minutes, we will follow the attacker through five doors into Nusrat's life: WhatsApp, Gmail, Facebook, Instagram and Telegram. At each door we ask one question: was the lock strong, or did it leak?

Slide 2: In 81 minutes, one shared code cost Nusrat four accounts (about 1 minute)

Here is how Nusrat's night unfolded. *[Follow the timeline left to right.]*

11:49: she shares the code. She never set a two-step PIN, so the attacker takes her WhatsApp.

11:58: her friends start receiving messages from "Nusrat" asking them to send money by bKash urgently.

12:20: the attacker sends her a fake "Facebook copyright violation" link. She logs in on the fake page. She uses the same password for Gmail.

12:35: from her Gmail, the attacker resets her Facebook password and replaces her recovery email and phone number.

1:10: Instagram falls next, and her private messages are used to blackmail her.

81 minutes. Four accounts.

[Point to the bottom row.] And this is not just a story. In the Cyber Crime Awareness Foundation's 2024 study, account hacking was the number one cybercrime. About 59 percent of victims were women, 78.78 percent were aged 18 to 30, and only 12 percent went to the police.

[Member 2] will explain how we investigated each door.

MEMBER 2 – [Your Name]

Slide 3: We retraced the attacker's path and tested every lock (about 1 minute)

Thank you, [Member 1]. So we became investigators. We retraced the attacker's path through Nusrat's five accounts, the five doors you see at the top. Four were breached that night. The fifth, Telegram, was not attacked, but as you will see, it was not as safe as she believed.

For each platform, we opened the security settings on our own phones and checked them against official help pages and recent news. Then we tested each door with the same five lock tests. *[Point to the bottom row.]*

Can a stolen password alone get in? Can the company read my messages? Will it warn me about an attack? Can a hacker lock me out permanently? And can strangers find and contact me?

Each test is scored from 1 to 5, so a perfect door scores 25.

Let us start where the attack started: WhatsApp, at 11:49.

Slide 4: Door 1 – WhatsApp: the strongest lock, opened from the inside (about 1 minute)

Door one: WhatsApp, 11:49 PM. Score: 20 out of 25, joint highest.

[Story banner.] Here is the irony. WhatsApp has the strongest lock of all five. Every personal chat and call is end-to-end encrypted by default; not even WhatsApp can read them. That encryption never broke. Nusrat opened the door from the inside by handing over the code.

[Left list.] And WhatsApp offered more locks she never used: a six-digit two-step PIN, which would have stopped this attack completely; passkeys; passkey-locked backups since October 2025; and since January 2026, Strict Account Settings, a one-tap lockdown.

[Fix card.] Our recommendation: when someone requests a registration code, show a clear warning in Bangla, "Never share this code, even with friends", and make the two-step PIN mandatory.

With WhatsApp gone, the attacker moved to the master key. [Member 3]?

MEMBER 3 – [Your Name]

Slide 5: Door 2 – Gmail: the master key that opens every other door (about 1 minute)

Thank you, [Member 2]. Door two: Gmail, 12:20 AM. Score: 20 out of 25, tied for first.

[Story banner.] The attacker sent Nusrat a fake "copyright violation" page. She typed in her password, the same password she used for Gmail. She had never switched on 2-Step Verification.

[Left list.] That is painful, because Gmail has the best login protection we tested. Passkeys have been the default sign-in for personal accounts since October 2023. There is 2-Step Verification, a Security Checkup and critical sign-in alerts, and Google says its filters block over 99.9 percent of phishing and malware.

Why does Gmail matter so much? Because every "forgot password" link, for Facebook, Instagram, even bKash, goes to your email. Gmail is the master key.

[Fix card.] We recommend optional end-to-end encryption for personal mail, and special warnings for emails imitating bKash, Nagad or local banks.

And with the master key, the next door opened easily.

Slide 6: Door 3 – Facebook: strong locks, but the attacker changed them (about 1 minute)

Door three: Facebook, 12:35 AM. Score: 16 out of 25.

[Story banner.] With Gmail in hand, the attacker clicked "forgot password" on Facebook, received the reset link, and then did what every attacker does first: replaced Nusrat's recovery email and phone number. By morning, Facebook believed he was Nusrat.

[Left list.] Facebook actually has good locks: two-factor authentication through an app, SMS or a security key; login alerts; passkeys since June 2025; profile lock; and encrypted Messenger chats. But a lock only helps if it is switched on.

[Score panel.] Its weakest score is recovery safety: just 2 out of 5. Changing recovery details takes minutes; getting your account back can take weeks.

[Fix card.] We propose a 48-hour cool-down before recovery details can change, with alerts sent to the old email and phone, and real human support in Bangla.

[Member 4] will take us to the next door.

MEMBER 4 – [Your Name]

Slide 7: Door 4 – Instagram: her private messages were never truly private (about 1 minute)

Thank you, [Member 3]. Door four: Instagram, 1:10 AM. Score: 16 out of 25.

[Story banner.] Nusrat's Instagram was linked to her Facebook through Meta's Accounts Center, so it fell next. The attacker scrolled through her private messages, and those messages became blackmail material. This is how sextortion usually begins.

[Score panel.] Look at encryption: 1 out of 5. Instagram DMs were never end-to-end encrypted by default, and on 8 May 2026 Meta removed even the optional encrypted chats.

[Left list.] To be fair, Instagram has useful tools. One of our favourites is "Emails from Instagram": it lists every genuine security email Instagram sent you in the last 14 days. Remember the fake "copyright" email at 12:20? This feature would have exposed it.

[Fix card.] Our recommendation: bring back encrypted DMs as the default, and blur unsolicited explicit images for every user, not just teenagers.

Slide 8: Door 5 – Telegram: the door that only looks locked (about 1 minute)

Door five: Telegram. It was not attacked that night, but it was the door Nusrat trusted most, and it scored the lowest: 14 out of 25.

[Story banner.] Like many students, she believed Telegram was the "private" app. Her class group lived there. But ordinary Telegram chats, groups and channels are not end-to-end encrypted; they are stored on Telegram's servers. Only "Secret Chats" are fully encrypted, and you have to start them manually. Since September 2024, Telegram also says it will share the IP addresses and phone numbers of rule-breaking users with authorities on valid legal requests.

[Left list.] It does offer good tools: a two-step cloud password, a passcode lock, control over active sessions, and the option to hide your phone number.

[Fix card.] We recommend end-to-end encryption by default for one-to-one chats, and a prompt to set up two-step verification at sign-up.

[Member 5] will now rewind the night.

MEMBER 5 – [Your Name]

Slide 9: Rewind: which locks held? WhatsApp is the most secure by default (about 1 minute)

Thank you, [Member 4]. Let us rewind the night and put all five doors side by side.

[Chart.] WhatsApp and Gmail tie at the top with 20. Facebook and Instagram score 16, and Telegram is last with 14.

So which platform is the most secure for an ordinary person? We chose WhatsApp. Its strongest lock, end-to-end encryption, is switched on from the moment you install it, with no setup. In Nusrat's story, it was never broken; she opened it herself.

[Second card.] Gmail deserves the most protection, because it is the master key. Once it fell at 12:20, Facebook and Instagram followed within an hour.

[Third card.] And Telegram reminds us that a reputation for privacy is not the same as privacy by default.

But here is the most important finding of our whole investigation: in Nusrat's story, not a single lock was broken. Every door was opened because a lock was switched off, or a code was handed over.

Slide 10: Replay the night: Same night. Different ending. (about 1 minute)

So let us replay the night, with five small changes. *[Move across the cards.]*

11:49: if Nusrat had a two-step PIN on WhatsApp, the code alone would be useless. Hijack blocked.

11:49: if she knew never to share a code, not even with a friend, the door stays shut.

12:20: with a unique password and a passkey on Gmail, the fake page steals nothing. The master key is safe.

12:35: with up-to-date recovery details and login alerts on, the takeover is spotted immediately.

1:10: by logging out of devices she does not recognise, she removes the attacker.

Each of these takes less than five minutes. All of them already exist in the apps on your phone.

[Pause. Read the last line slowly.] 11:47 PM. The message arrives. This time, Nusrat says no.

The most secure app is the one you set up. Thank you; we are happy to take your questions.